

Rear View Mirrors — Exterior



Special Tool(s)

 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

LH and RH exterior mirror glass movement is controlled by the exterior mirror control switch and the LH and RH exterior mirror motors. Adjusting the exterior mirror control switch to the LH or RH position determines which exterior mirror motor to control.

The exterior mirrors use a jumper harness between the vehicle wire harness connector and the exterior mirror motor and exterior mirror glass. The exterior mirror jumper harness is integral to the exterior mirror. Before replacing an exterior mirror or motor, inspect the jumper harness for open and shorted circuits or damaged and pushed-out pins. If a concern with the exterior mirror jumper harness exists, attempt to repair the jumper harness.

If equipped with power folding mirrors, the center position of the exterior mirror control switch controls the power folding mirrors.

If equipped with power telescoping mirrors, the telescoping mirror control switch, part of the exterior mirror control switch, allows the power telescoping mirrors to be extended or retracted.

Vehicles equipped with memory exterior mirrors are capable of setting DTCs for the mirror circuits.

Power Exterior Mirrors — Without Memory

Exterior mirror glass logic is the selected request from the exterior mirror control switch. When the switch is adjusted to the left, right, up or down position, voltage and ground is supplied to the selected exterior mirror motor to move the exterior mirror glass to the desired position.

Memory Exterior Mirrors

LH and RH exterior mirror glass movement is controlled by the exterior mirror control switch, Driver Seat Module (DSM) and the LH and RH exterior mirror motors. Adjusting the switch to the LH or RH position determines which exterior mirror

motor to control. The switch sends voltage and ground input signals to the **DSM**. The **DSM** provides voltage and ground to the appropriate exterior mirror motor to move the selected exterior mirror glass to the desired position.

The memory function recalls the preferred position of the exterior mirrors when the gear selector is in PARK or NEUTRAL or the Remote Keyless Entry (RKE) transmitter is activated. The **DSM** constantly tracks the exterior mirror motor position using potentiometers located internal to the exterior mirror motors. When a memory position is recalled, the **DSM** powers the exterior mirror motors simultaneously and monitors the potentiometer circuits. When the stored memory position is reached, the **DSM** removes the voltage and ground from the motors. If the exterior mirror control switch is operated during a memory recall, the **DSM** stops the recall and responds to the exterior mirror control switch command. The **DSM** is capable of setting a DTC for the memory mirror system when a system concern exists.

For information on the **DSM** and memory position programming, refer to Section 501-10.

Heated Exterior Mirror

The heated exterior mirror glass and if equipped, the heated spotter mirror glass operates when the defrost system is activated at one of the three following locations: HVAC module, FCIM or FDIM (8- inch [203mm] touch screen). The defrost system can also be activated with a remote start (if equipped). The LH and RH heated exterior mirror glass voltage is supplied by the rear window defrost relay through Battery Junction Box (BJB) fuse 32 (40A).

The heated mirrors are heated for a period of 10 minutes when the engine is running and the defrost system is activated.

On vehicles equipped with a power sliding rear window with rear defrost, the defrost switch LED does not illuminate unless the rear sliding window glass is in the fully CLOSED position.

On vehicles equipped with a LH auto-dimming exterior mirror, the glass is thicker than the standard mirror glass. As a result, it takes longer for a LH auto-dimming exterior mirror to defrost than a RH heated exterior mirror.

LH Auto-Dimming Exterior Mirror

The LH exterior mirror is equipped with an auto-dimming feature. This feature automatically reduces the glare caused by headlamps reflecting in the mirror. The interior auto-dimming mirror controls the reflectance level of the LH exterior mirror. The reflectance level of the mirror glass is variable and depends on the amount of rear glare in relation to ambient light conditions in front of the interior mirror. To provide increased visibility when backing up, the interior mirror glass and LH exterior mirror glass automatically return to a high reflectance mode whenever the selector lever is placed in REVERSE.

For information and diagnostics, refer to Rear View Mirrors — Interior in this section.

Power Folding Mirrors (With Or Without Telescoping Mirrors)

NOTE: *The power folding mirrors may need to be synchronized when the RH, LH or both power folding mirrors are folded manually, and then folded inward using the power folding mirror control switch. Refer to Exterior Mirror Motor Synchronization — Power Folding in this section.*

NOTE: *Before running a self-test diagnostic or retrieving DTCs, the power fold switch must be inactive for at least 30 seconds. Retrieving DTCs before 30 seconds results in false DTCs.*

The power folding exterior mirror feature allows both power exterior rear view mirrors to be folded or unfolded using the exterior mirror control switch. On the vehicles equipped with the power folding exterior mirrors without power telescoping mirrors, the power folding exterior mirrors are controlled by the **DSM**. When the exterior mirror control switch is set to the center position and pressed rearward to the FOLD/UNFOLD position, the exterior mirror control switch sends a signal to the **DSM** which then supplies power and ground to the exterior mirrors to fold or unfold the mirrors. On the vehicles equipped with the power folding exterior mirrors and power telescoping mirrors, the power folding exterior mirrors are controlled by the power mirror fold relay. When the exterior mirror control switch is set to the center position and pressed rearward to the FOLD/UNFOLD position, the exterior mirror control switch sends a ground signal to the power mirror fold relay. The power mirror fold relay activate the power folding mirrors in the opposite direction of the last activation. The power mirror fold relay controls movement by reversing the polarity of the voltage and ground it supplies to the power folding mirror motors. The RH and LH power folding mirrors fold or unfold at the same time and cannot be controlled independently. The power folding mirrors contain a drive motor that is not replaceable. If a power folding mirror drive motor is found to be inoperative, a new power folding exterior rear view mirror assembly must be installed.

If the power folding mirrors are folded or unfolded continuously (6 to 10 cycles), the power lockout feature disables the system for approximately 3-10 minutes to prevent damage to the power folding mirror drive motors. After 3-10 minutes have elapsed, normal operation resumes. Before running a self-test or retrieving DTCs, the power fold switch must be inactive for at least 30 seconds.

Power Telescoping Mirrors

The power telescoping mirror feature allows the LH and RH power exterior mirrors to be electronically extended or retracted using the telescoping rear view mirror control switch. When the telescoping rear view mirror control switch is pressed to the outward position, the exterior rear view mirrors will extend. When the telescoping rear view mirror control switch is pressed to the inward position, the exterior rear view mirrors retract. The RH and LH power telescoping mirrors extend and retract at the same time and cannot be controlled independently. The power telescoping mirrors contain a drive motor which is not replaceable.

Turn Signal Mirror

The LH and RH exterior mirrors may be equipped with a mirror-mounted turn signal assembly. For repair procedures and information on the mirror-mounted turn signals, refer to Section 417-01.

Puddle/Approach Lamps

The LH and RH exterior mirrors may be equipped with approach or puddle lamps, which will illuminate when the interior lights are turned on by opening a door or by pressing the unlock button on the transmitter, if equipped. For repair procedures and information on the mirror-mounted puddle lamps, refer to Section 417-02.

Inspection and Verification

NOTE: *Clean the entire mirror assembly and glass to assist in verification of impact damage.*

1. Verify the customer concern by operating the power exterior mirrors or memory mirrors.
2. Visually inspect for obvious signs of mechanical and electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Exterior mirror • Exterior mirror control switch • Power telescoping mirror control switch • Exterior mirror glass • Exterior mirror cover	• Body Control Module (BCM) fuses: ▪ 7 (7.5A) ▪ 15 (15A) • Battery Junction Box (BJB) fuses: ▪ 32 (40A) ▪ 56 (10A) ▪ 44 (15A) • Driver Seat Module (DSM) • Exterior mirror control switch • Power telescoping mirror control switch • Exterior mirror glass motor • Exterior mirror power telescoping motor (part of the exterior mirror) • Exterior mirror power folding motor (part of the exterior mirror) • Exterior mirror glass • Exterior spotter mirror glass • Rear window defrost relay • Power mirror fold relay • HVAC — Dual Automatic Temperature Control (DATC) Module • HVAC — Electronic Manual Temperature Control (EMTC) Module

Mechanical	Electrical
	• HVAC — Single Automatic Temperature Control • FCIM • Loose or corroded connections • Wiring harness

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The LED prove-out confirms power and ground from the are provided to the DLC .*

If the scan tool does not communicate with the Vehicle Communication Module (VCM) :

- check the connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00 to diagnose no response from the DSM .

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, proceed with module self-test diagnostics.

8. Carry out the self-test diagnostics for the BCM , and DSM .

9. If the DTCs retrieved are related to the concern, go to DTC Charts.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

11. If no DTCs related to the concern are retrieved, GO to [Symptom Chart — Exterior Mirrors](#) or GO to [Symptom Chart — NVH](#).

DTC Charts

Driver Seat Module (DSM) DTC Chart

DTC	Description	Action
B112B	Mirror Power Fold Switch Input Short to Battery	GO to Pinpoint Test E.
B106E	Solid State Driver Disabled Due to Short Circuit	GO to Pinpoint Test G.
B106F	Module Disabled Due to External Fault	GO to Pinpoint Test G.
B1667	Mirror Driver Up/Down Motor Stalled	GO to Pinpoint Test D.
B1668	Mirror Driver Right/Left Motor Stalled	GO to Pinpoint Test D.
B1669	Mirror Passenger Up/Down Motor Stalled	GO to Pinpoint Test D.
B1670	Mirror Passenger Right/Left Motor Stalled	GO to Pinpoint Test D.
B1733	Mirror Driver Vertical Switch Circuit Failure	GO to Pinpoint Test D.

DTC	Description	Action
B1737	Mirror Driver Horizontal Switch Circuit Failure	GO to Pinpoint Test D.
B1741	Mirror Passenger Vertical Switch Circuit Failure	GO to Pinpoint Test D.
B1745	Mirror Passenger Horizontal Switch Circuit Failure	GO to Pinpoint Test D.
B2223	Mirror Driver Drive Circuit Failure	GO to Pinpoint Test D.
B2224	Mirror Passenger Drive Circuit Failure	GO to Pinpoint Test D.
B2312	Mirror Passenger Horizontal Feedback Potentiometer Circuit Failure	GO to Pinpoint Test D.
B2314	Mirror Passenger Horizontal Feedback Potentiometer Circuit Short to Battery	GO to Pinpoint Test D.
B2316	Mirror Passenger Vertical Feedback Potentiometer Circuit Failure	GO to Pinpoint Test D.
B2318	Mirror Passenger Vertical Feedback Potentiometer Circuit Short to Battery	GO to Pinpoint Test D.
B2320	Mirror Driver Horizontal Feedback Potentiometer Circuit Failure	GO to Pinpoint Test D.
B2322	Mirror Driver Horizontal Feedback Potentiometer Circuit Short to Battery	GO to Pinpoint Test D.
B2324	Mirror Driver Vertical Feedback Potentiometer Circuit Failure	GO to Pinpoint Test D.
B2326	Mirror Driver Vertical Feedback Potentiometer Circuit Short to Battery	GO to Pinpoint Test D.
All other DTCs	—	REFER to Section 501-10.

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. ~~CMDTCs~~ have an additional 2-character DTC status code suffix to assist in determining DTC history.

HVAC Module DTC Chart

DTC	Description	Action
B1C83:11	Rear Defog Relay: Circuit Short to Ground	GO to Pinpoint Test F.
B1C83:12	Rear Defog Relay: Circuit Short to Battery	GO to Pinpoint Test F.
B1C83:13	Rear Defog Relay: Circuit Open	GO to Pinpoint Test F.
All other DTCs	—	REFER to Section 412-00.

Symptom Chart — Exterior Mirrors

Symptom Chart — Exterior Mirrors		
Condition	Possible Causes	Action
Mirror folds inward due to wind pressure	- Power folding mirrors are not synchronized - Mirror structural bracket is damaged	- SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. - INSPECT the mirror for structural damage. If necessary, INSTALL a new mirror. REFER to Exterior Mirror in this section.

• The mirrors are inoperative — turn indicator	• Turn signal module • Fuse • Wiring, terminals or connectors • Exterior mirrors	• REFER to Section 417-01 for diagnosis and individual component removal and installation.
• The mirrors are inoperative — side marker	• Fuse • Wiring, terminals or connectors • Exterior mirrors	• REFER to Section 417-01 for diagnosis and individual component removal and installation.
• The mirrors are inoperative — puddle/approach lamp	• Puddle lamp module • Fuse • Wiring, terminals or connectors • Exterior mirrors	• REFER to Section 417-02 for diagnosis and individual component removal installation.
• The mirrors are inoperative — non-memory mirrors	• Fuse • Wiring, terminals or connectors • Exterior mirror control switch	• GO to Pinpoint Test A.
• A single mirror is inoperative — non-memory mirrors	• Wiring, terminals or connectors • Exterior mirror control switch • Exterior mirror motor • Exterior mirror	• GO to Pinpoint Test B.
• A single mirror does not operate correctly — non-memory mirrors	• Wiring, terminals or connectors • Exterior mirror motor • Exterior mirror control switch • Exterior mirror	• GO to Pinpoint Test C.

• The memory mirror is inoperative/does not operate correctly	• Fuse • Wiring, terminals or connectors • Exterior mirror control switch • Exterior mirror motor • Exterior mirror • DSM	• GO to Pinpoint Test D.
• The memory mirror is inoperative — both mirrors	• Fuse • Wiring, terminals or connectors • Exterior mirror control switch • DSM	• GO to Pinpoint Test D.
• The power telescoping mirrors do not operate/do not operate correctly	• Fuse • Wiring, terminals or connectors • Power telescoping mirror control switch • Power telescoping mirror	• GO to Pinpoint Test I.
• The power folding mirrors do not operate/do not operate correctly	• The power folding mirror motor lockout feature is active	• WAIT 3 minutes with the engine running or 10 minutes with the engine not running for the power folding mirror lockout feature to reset.
• The power folding mirrors do not operate/do not operate correctly (without power telescoping mirrors)	• The power folding mirror motor lockout feature is active • Power folding mirrors not synchronized • Binding or obstructed power folding mirror • Wiring, terminals or connectors	• GO to Pinpoint Test E.

	• Exterior mirror control switch • Power folding mirror • DSM	
• The power folding mirrors do not operate/do not operate correctly (with power telescoping mirrors)	• The power folding mirror motor lockout feature is active • Power folding mirrors not synchronized • Binding or obstructed power folding mirror • Fuse • Wiring, terminals or connectors • Exterior mirror control switch • Power mirror fold relay • Power folding mirror	• GO to Pinpoint Test J.
• The heated exterior mirror is inoperative — one or both mirrors	• Fuse • Wiring, terminals or connectors • Exterior mirror glass • Exterior spotter mirror glass • Exterior mirror • Rear window defrost relay • HVAC module • ECM	• GO to Pinpoint Test F.
• The heated exterior mirrors do not operate from voice command	• Voice recognition system	• VERIFY the heated exterior mirrors operation using the instrument panel defrost switch. If the heated exterior mirrors are still inoperative, GO to Pinpoint Test F. If the

		heated exterior mirrors operate, Refer to Section 415-00D or Section 415-00E.
The heated exterior mirrors do not operate from touchscreen command	Touchscreen	VERIFY the heated exterior mirrors operation using the instrument panel defrost switch. If the heated exterior mirrors are still inoperative, GO to Pinpoint Test F. If the heated exterior mirrors operate, Refer to Section 415-00D or Section 415-00E.
The heated exterior mirrors do not operate correctly — RH and LH do not defrost/heat at the same rate	Normal operation of the heated exterior mirror	No action required. System operating correctly at this time.
The LH exterior auto-dimming mirror does not operate correctly	- Fuse - Wiring, terminals or connectors - Interior auto-dimming mirror - LH exterior mirror glass - LH exterior mirror	- If the interior auto-dimming mirror is not operating correctly, GO to Symptom Chart — Interior Mirror. - If the interior auto-dimming mirror operates correctly and the LH exterior auto-dimming mirror does not operate correctly, GO to Pinpoint Test H.

Symptom Chart — NVH

Symptom Chart — NVH

Condition	Possible Causes	Action
NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.		
Mirror vibrates/loose	- Exterior mirror motor screws loose - Exterior mirror mounting nuts loose - Mirror glass loose	- REMOVE mirror glass and TIGHTEN the 3 exterior mirror motor screws. - TIGHTEN the exterior mirror mounting nuts. REFER to Specifications in this section. - PRESS the center of the exterior mirror glass up, down, left and right to make sure that the exterior mirror glass is seated correctly. If the exterior mirror glass is still loose, REMOVE the exterior mirror glass and INSPECT the exterior mirror backing plate for damage. If the exterior mirror backing plate is damaged, INSTALL a new exterior mirror glass. REFER to Exterior Mirror Glass — Aero in this section.

	• Aftermarket air deflector/stone shields • Power folding mirrors are not synchronized • Exterior mirror is structurally damaged • Exterior mirror dampener springs are damaged or missing (LH only) • Exterior mirror motor is damaged	• If possible, REMOVE aftermarket air deflector/stone shield, then ROAD TEST the vehicle. If concern is no longer present, ADVISE customer that aftermarket components were causing undesired vibration. • SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. • INSPECT the exterior mirror for structural damage. If necessary, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. • REMOVE the exterior mirror glass. REFER to Exterior Mirror Glass — Aero in this section. INSPECT the exterior mirror glass for damaged or missing mirror exterior mirror dampener springs. If necessary, INSTALL a new exterior mirror glass. REFER to Exterior Mirror Glass — Aero in this section. • REMOVE the exterior mirror glass. REFER to Exterior Mirror Glass — Aero in this section. INSPECT the exterior mirror motor for damage. If necessary, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section.
• Wind noise	• Foam gasket between exterior mirror housing and door is misaligned or damaged • Exterior mirror is not correctly fitted to the door • Exterior mirror cover • Damaged exterior mirror heat stakes	• VERIFY gasket is in good condition. If necessary, REPOSITION foam gasket between exterior rear view mirror housing and door frame or REPAIR the foam gasket as needed. The foam gasket is not serviceable and if it cannot be REPAIRED, a new exterior mirror needs to be installed. REFER to Exterior Mirror in this section. • VERIFY that there are no gaps between the exterior mirror and the door. If necessary, LOOSEN the exterior mirror nuts and REPOSITION the exterior mirror. • VERIFY that the exterior mirror cover is not loose or broken. If necessary, INSTALL a new exterior mirror cover. REFER to Exterior Mirror Cover — Aero in this section. • INSTALL a new exterior mirror. REFER to Exterior Mirror in this section.

Pinpoint Tests

Pinpoint Test A: The Mirrors are Inoperative — Non-Memory Mirrors

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

Normal Operation

The exterior mirror control switch receives voltage at all times from the Body Control Module (BCM) . The exterior mirror control switch has only one ground circuit. The exterior mirrors share a common circuit for both up/down and right/left movement.

This pinpoint test is intended to diagnose the following:

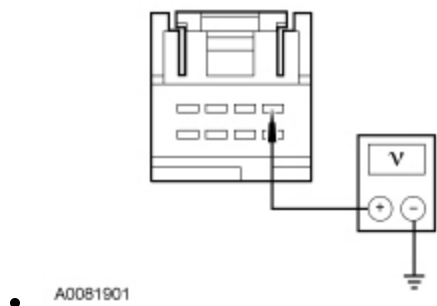
- Fuse
- Wiring, terminals or connectors
- Exterior mirror control switch

PINPOINT TEST A : THE MIRRORS ARE INOPERATIVE — NON-MEMORY MIRRORS

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK VOLTAGE TO THE EXTERIOR MIRROR CONTROL SWITCH

- Disconnect: Exterior Mirror Control Switch C527 .
- Measure the voltage between exterior mirror control switch [C527](#) Pin 1, circuit SBP07 (WH/RD), harness side and ground.

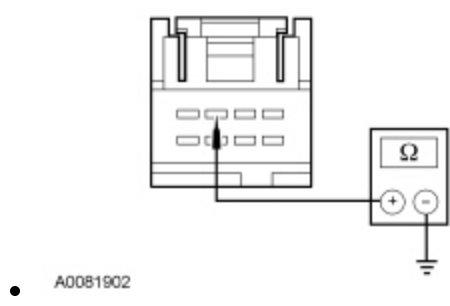


Is the voltage greater than 11 volts?

Yes	GO to A2 .
No	VERIFY that fuse 7 (7.5A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

A2 CHECK FOR GROUND TO THE EXTERIOR MIRROR CONTROL SWITCH

- Measure the resistance between exterior mirror control switch [C527](#) Pin 3, circuit GD134 (BK/VT), harness side and ground.

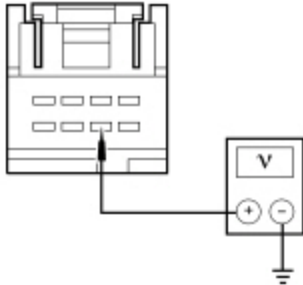


Is the resistance less than 3 ohms?

Yes	GO to A3 .
No	REPAIR the circuit. TEST the system for normal operation.

A3 CHECK THE EXTERIOR REAR VIEW MIRROR COMMON CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: Exterior Mirror LH [C521](#).
- Disconnect: Exterior Mirror RH [C615](#).
- Ignition ON.
- Measure the voltage between exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side and ground.



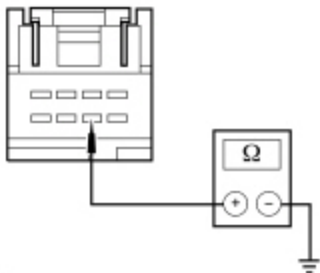
• A0073142

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to A4 .

A4 CHECK THE EXTERIOR REAR VIEW MIRROR COMMON CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side and ground.



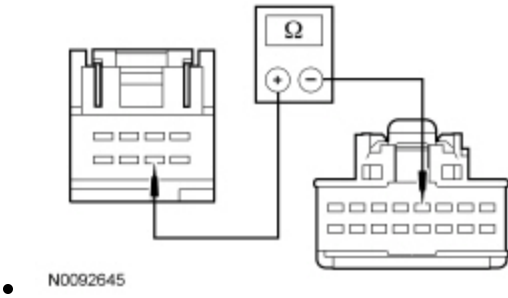
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Is the resistance greater than 10,000 ohms?

Yes	GO to A5 .
No	REPAIR the circuit. TEST the system for normal operation.

A5 CHECK THE EXTERIOR REAR VIEW MIRROR COMMON CIRCUIT FOR AN OPEN

- Measure the resistance between exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side and LH exterior mirror [C521](#) Pin 4, circuit CPM23 (GY), harness side.



Is the resistance less than 3 ohms?

Yes	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. TEST the system for normal operation. mm (in)
No	REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test B: A Single Mirror is Inoperative — Non-Memory Mirrors

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

Normal Operation

The exterior mirror control switch uses a common circuit for both exterior mirror down/up and left/right movement.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Exterior mirror control switch
- Exterior mirror motor
- Exterior mirror

PINPOINT TEST B : A SINGLE MIRROR IS INOPERATIVE — NON-MEMORY MIRRORS

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

B1 CHECK THE EXTERIOR MIRROR CONTROL SWITCH

- Ignition OFF.
- Disconnect: Exterior Mirror Control Switch [C527](#).
- Carry out the Exterior Mirror Control Switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

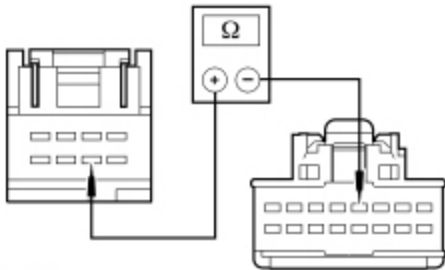
Did the exterior mirror control switch pass the component test?

Yes	GO to B2 .
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No	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. TEST the system for normal operation.
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B2 CHECK THE INOPERATIVE EXTERIOR REAR VIEW MIRROR COMMON CIRCUIT FOR AN OPEN

- Disconnect: LH Exterior Mirror [C521](#) or RH Exterior Mirror [C615](#).
- Measure the resistance:
 - for LH inoperative mirror, between [C521](#) Pin 4, circuit CPM23 (GY), harness side and exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side.
 - for RH inoperative mirror, between [C615](#) Pin 4, circuit CPM23 (GY), harness side and exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side.



N0092645

Is the resistance less than 3 ohms?

Yes	CHECK the inoperative exterior mirror jumper harness between the vehicle harness and the exterior mirror motor for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test C: A Single Mirror Does Not Function With Switch Logic — Non-Memory Mirrors

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

Normal Operation

The exterior mirror control switch supplies voltage and ground to control LH and RH exterior mirror motor movement. Each mirror contains a horizontal and vertical motor. The exterior mirror control switch controls movement by reversing polarity of the voltage and ground circuits being supplied to each motor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Exterior mirror control switch
- Exterior mirror motor
- Exterior mirror

PINPOINT TEST C : A SINGLE MIRROR DOES NOT FUNCTION WITH SWITCH LOGIC — NON-MEMORY MIRRORS

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

C1 CHECK THE MIRROR MOVEMENT (LEFT/RIGHT)

- Operate the LH and RH exterior mirrors in the left and right direction using the exterior mirror control switch.

Do both exterior mirrors move left and right?

Yes	GO to C2 .
No	GO to C3 .

C2 CHECK THE MIRROR MOVEMENT (UP/DOWN)

- Operate the LH and RH exterior mirrors in the up and down direction using the exterior mirror control switch.

Do both exterior mirrors move up and down?

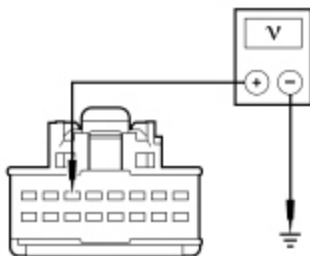
Yes	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. TEST the system for normal operation.
No	GO to C4 .

C3 CHECK THE VOLTAGE TO THE INOPERATIVE MIRROR LEFT/RIGHT CIRCUITS

- Disconnect: Inoperative Exterior Mirror [C521](#) (LH) or [C615](#) (RH) .
- **NOTE:** *Position the exterior mirror control switch to the direction of the inoperative mirror.*

While operating the exterior mirror control switch in the left direction, measure the voltage between ground and:

- **LH exterior mirror:** [C521](#) Pin 6, circuit CPM16 (BN/BU), harness side.
- **RH exterior mirror:** [C615](#) Pin 6, circuit CPM20 (BN/WH), harness side.



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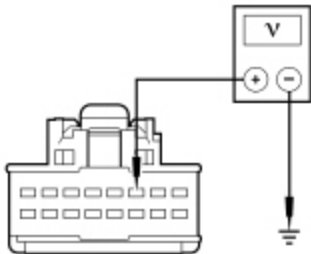
Is the voltage greater than 11 volts?

Yes	CHECK the inoperative exterior mirror jumper harness between the vehicle harness and the exterior mirror motor for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a
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	new exterior mirror motor. REFER to Exterior Mirror Motor in this section. TEST the system for normal operation.
No	GO to C5 .

C4 CHECK THE VOLTAGE TO THE INOPERATIVE MIRROR UP/DOWN CIRCUITS

- Disconnect: Inoperative Exterior Mirror [C521](#) (LH) or [C615](#) (RH) .
 - **NOTE:** *Position the exterior mirror control switch to the direction of the inoperative mirror.*
- While operating the exterior mirror control switch in the upward direction, measure the voltage between ground and:
- **LH exterior mirror:** [C521](#) Pin 3, circuit CPM17 (BU/GN), harness side.
 - **RH exterior mirror:** [C615](#) Pin 3, circuit CPM21 (YE/VT), harness side.



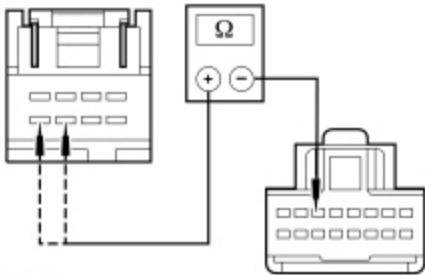
N0092647

Is the voltage greater than 11 volts?

Yes	CHECK the inoperative exterior mirror jumper harness between the vehicle harness and the exterior mirror motor for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. TEST the system for normal operation.
No	GO to C6 .

C5 CHECK THE MIRROR CONTROL LEFT/RIGHT CIRCUITS FOR AN OPEN

- Disconnect: Exterior Mirror Control Switch [C527](#).
- **LH exterior mirror:** Measure the resistance between exterior mirror control switch [C527](#) Pin 8, circuit CPM16 (BN/BU), harness side and LH exterior mirror [C521](#) Pin 6, circuit CPM16 (BN/BU), harness side.
- **RH exterior mirror:** Measure the resistance between exterior mirror control switch [C527](#) Pin 7, circuit CPM20 (BN/WH), harness side and RH exterior mirror [C615](#) Pin 6, circuit CPM20 (BN/WH), harness side.



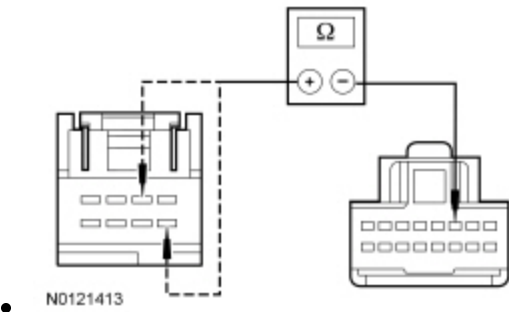
N0121411

Is the resistance less than 3 ohms?

Yes	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

C6 CHECK THE MIRROR CONTROL UP/DOWN CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Exterior Mirror Control Switch [C527](#).
- **LH exterior mirror:** Measure the resistance between exterior mirror control switch [C527](#) Pin 5, circuit CPM17 (BU/GN), harness side and LH exterior mirror [C521](#) Pin 3, circuit CPM17 (BU/GN), harness side.
- **RH exterior mirror:** Measure the resistance between exterior mirror control switch [C527](#) Pin 2, circuit CPM21 (YE/VT), harness side and RH exterior mirror [C615](#) Pin 3, circuit CPM21 (YE/VT), harness side.



Is the resistance less than 3 ohms?

Yes	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test D: The Memory Mirror is Inoperative

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

Normal Operation

The exterior mirror control switch sends voltage and ground inputs to the Driver Seat Module (DSM) which controls mirror movement by providing the appropriate voltage and ground polarity to the LH/RH exterior mirror up/down and left/right motors. The LH and RH exterior mirrors each contain potentiometers for up/down and left/right position status. The DSM supplies the LH and RH exterior mirror feedback potentiometers a reference voltage and reference ground. The DSM uses these feedback signals to store/recall mirror positions.

DTC Description	Fault Trigger Conditions
• B1667 — Mirror Driver Up/Down Motor Stalled • B1669 — Mirror Passenger Up/Down Motor Stalled	During self-test, the <u>DSM</u> attempts to operate the exterior mirror motor and uses the motor's position sensor to monitor movement. If no motor movement is sensed, the <u>DSM</u> disables the output which operates the motor after 250 milliseconds and the DTC is set. The <u>DSM</u> retries motor operation on the next activation of the exterior mirror control switch.
• B1668 — Mirror Driver Right/Left Motor Stalled • B1670 — Mirror Passenger Right/Left Motor Stalled	During self-test, the <u>DSM</u> attempts to operate the exterior mirror motor and uses the motor's position sensor to monitor movement. If no motor movement is sensed, the <u>DSM</u> disables the output which operates the motor after 250 milliseconds and the DTC is set. The <u>DSM</u> retries motor operation on the next activation of the exterior mirror control switch.
• B1733 — Mirror Driver Vertical Switch Circuit Failure • B1741 — Mirror Passenger Vertical Switch Circuit Failure	During self-test, if a short to ground or short to battery fault is sensed on the switch circuit, the DTC is set. During normal operation, if a fault is sensed on the circuit for greater than 2 minutes, the DTC is set as continuous. With the DTC set, any input signal on the circuit is ignored.
• B1737 — Mirror Driver Horizontal Switch Circuit Failure • B1745 — Mirror Passenger Horizontal Switch Circuit Failure	During self-test, if a short to ground or short to battery fault is sensed on the switch circuit, the DTC is set. During normal operation , if a fault is sensed on the circuit for greater than 2 minutes, the DTC is set as continuous. With the DTC set, any input signal on the circuit is ignored.
• B2223 — Mirror Driver Drive Circuit Failure • B2224 — Mirror Passenger Drive Circuit Failure	Short to ground or battery detected by the <u>DSM</u> .
• B2312 — Mirror Passenger Horizontal Feedback Potentiometer Circuit Failure • B2320 — Mirror Driver Horizontal Feedback Potentiometer Circuit Failure	Open or short to ground detected by the <u>DSM</u> . When DTC is set <u>DSM</u> operates motor for 1 second then movement stops.
• B2314 — Mirror Passenger Horizontal Feedback Potentiometer Circuit Short to Battery • B2322 — Mirror Driver Horizontal Feedback Potentiometer Circuit Short to Battery	Short to battery detected by the <u>DSM</u> . When DTC is set <u>DSM</u> operates motor for 1 second then movement stops.
• B2316 — Mirror Passenger Vertical Feedback Potentiometer Circuit Failure • B2324 — Mirror Driver Vertical Feedback Potentiometer Circuit Failure	Open or short to ground detected by the <u>DSM</u> . When DTC is set <u>DSM</u> operates motor for 1 second then movement stops.

DTC Description	Fault Trigger Conditions
- B2318 — Mirror Passenger Vertical Feedback Potentiometer Circuit Short to Battery - B2326 — Mirror Driver Vertical Feedback Potentiometer Circuit Short to Battery	Short to battery detected by the DSM . When DTC is set DSM operates motor for 1 second then movement stops.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Exterior mirror control switch
- Exterior mirror motor
- Exterior mirror
- **DSM**

PINPOINT TEST D : THE MEMORY MIRROR IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
D1 RETRIEVE THE DTCS FROM THE DSM	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Self Test — DSM .	
Are any DTCs retrieved?	
Yes	If DTC B1667 or B1668 is present with other DTCs, DIAGNOSE the other DTCs first. If DTC B1667 or B1668 is present with no other DTCs, INSPECT the LH exterior mirror motor for foreign material, damage or a binding condition. If no condition is found, GO to D10 . If DTC B1669 or B1670 is present with other DTCs, DIAGNOSE the other DTCs first. If DTC B1669 or B1670 is present with no other DTCs, INSPECT the RH exterior mirror motor for foreign material, damage or a binding condition. If no condition is found, GO to D12 . For DTCs B1733, B1737, B1741 or B1745, GO to D14 . For DTC B2223, GO to D17 . For DTC B2224, GO to D19 . For DTCs B2312, B2316, B2320 or B2324, GO to D21 . For DTCs B2314, B2318, B2322 or B2326, GO to D24 . For all other DSM DTCs, REFER to Section 501-10.
No	GO to D2 .

D2 CHECK OPERATION OF THE MIRRORS FROM THE EXTERIOR MIRROR CONTROL SWITCH	
• Operate the mirrors using the exterior mirror control switch.	
Do the LH or RH mirrors operate using the exterior mirror control switch?	

Yes	If the LH or RH exterior mirror operates correctly in at least one direction GO to D9 .
No	If both the LH and RH exterior mirrors are inoperative, GO to D3 .

D3 CHECK THE EXTERIOR MIRROR CONTROL SWITCH

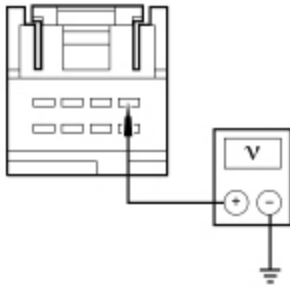
- Ignition OFF.
- Disconnect: Exterior Mirror Control Switch [C527](#).
- Carry out the Exterior Mirror Control Switch Component Test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Did the exterior mirror control switch pass the component test?

Yes	GO to D4 .
No	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. TEST the system for normal operation. mm (in)

D4 CHECK FOR VOLTAGE TO THE EXTERIOR MIRROR CONTROL SWITCH

- Measure the voltage between exterior mirror control switch [C527](#) Pin 1, circuit SBP07 (WH/RD), harness side and ground.

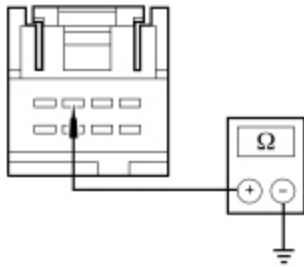


Is the voltage greater than 11 volts?

Yes	GO to D5 .
No	VERIFY that BCM fuse 7 (7.5A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

D5 CHECK FOR GROUND TO THE EXTERIOR MIRROR CONTROL SWITCH

- Measure the resistance between exterior mirror control switch [C527](#) Pin 3, circuit GD134 (BK/VT), harness side and ground.



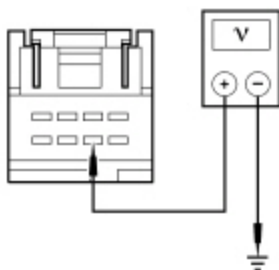
• A0081902

Is the resistance less than 3 ohms?

Yes	GO to D6 .
No	REPAIR the circuit. TEST the system for normal operation.

D6 CHECK THE MIRROR SWITCH COMMON CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: ~~DSM~~ [C341D](#) .
- Ignition ON.
- Measure the voltage between exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side and ground.



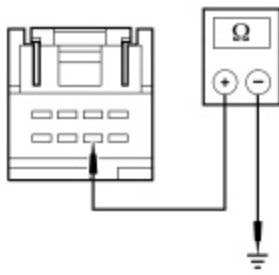
• N0077231

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to D7 .

D7 CHECK THE MIRROR SWITCH COMMON CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY) harness side and ground.



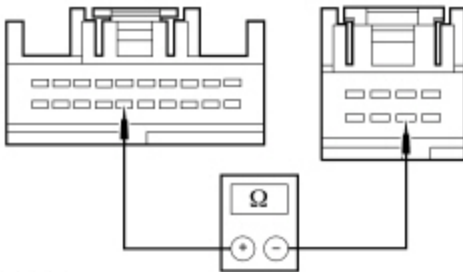
• N0077232

Is the resistance greater than 10,000 ohms?

Yes	GO to D8 .
No	REPAIR the circuit. TEST the system for normal operation.

D8 CHECK THE MIRROR SWITCH COMMON CIRCUIT FOR AN OPEN

- Measure the resistance between ~~DSM~~ [C341D](#) Pin 16, circuit RPM11 (YE/OG), harness side and exterior mirror control switch [C527](#) Pin 6, circuit CPM23 (GY), harness side.



• N0068340

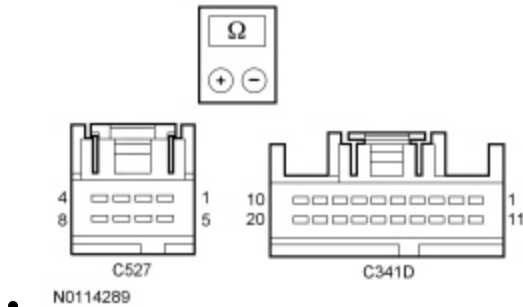
Is the resistance less than 3 ohms?

Yes	GO to D25 .
No	REPAIR the circuit. TEST system for normal operation.

D9 CHECK THE EXTERIOR MIRROR SWITCH CIRCUITS FOR AN OPEN

- Disconnect: Exterior Mirror Control Switch [C527](#).
- Disconnect: ~~DSM~~ [C341D](#) .
- Measure the resistance between exterior mirror control switch [C527](#), harness side and ~~DSM~~ [C341D](#), harness side using the following chart:

Exterior Mirror Switch	Circuit	DSM
C527 Pin 8	CPM16 (BN/BU)	C341D Pin 20
C527 Pin 5	CPM17 (BU/GN)	C341D Pin 19
C527 Pin 7	CPM20 (BN/WH)	C341D Pin 18
C527 Pin 2	CPM21 (YE/VT)	C341D Pin 17

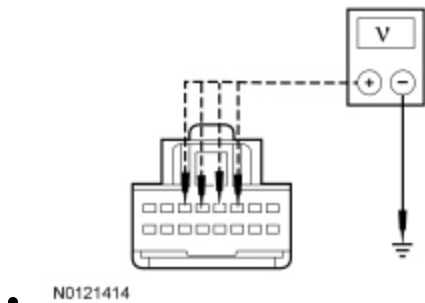


Are the resistances less than 3 ohms?

Yes	GO to D25 .
No	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D10 CHECK THE LH EXTERIOR MIRROR FOR VOLTAGE WHILE COMMANDING THE DSM PIDS

- Disconnect: LH Exterior Mirror [C521](#).
- Enter the following diagnostic mode on the scan tool: DataLogger — [DSM](#) .
- **DTC B1667:** Measure the voltage between ground and LH exterior mirror:
 - [C521](#) Pin 3, circuit CPM29 (GN), harness side while toggling the [DSM](#) PID Driver's Mirror Up (DR_UP) to On.
 - [C521](#) Pin 4, circuit CPM03 (BU/WH), harness side while toggling the [DSM](#) PID Driver's Mirror Down (DR_DOWN) to On.
- **DTC B1668:** Measure the voltage between ground and LH exterior mirror:
 - [C521](#) Pin 5, circuit CPM03 (BU/WH), harness side while toggling the [DSM](#) PID Driver's Mirror Left (DR_LEFT) to On.
 - [C521](#) Pin 6, circuit CPM28 (BU/BN), harness side while toggling the [DSM](#) PID Driver's Mirror Right (DR_RIGHT) to On.



Do the voltages momentarily change from 0 volts to greater than 10 volts when the active command is toggled ON?

Yes	CHECK the LH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new LH exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a
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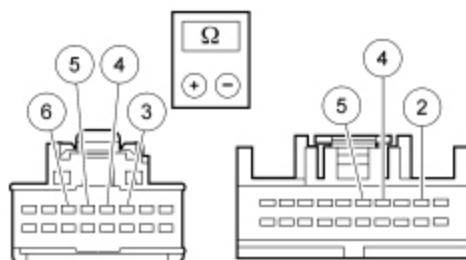
new LH exterior mirror motor. REFER to Exterior Mirror Motor in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

No GO to [D11](#).

D11 CHECK THE LH EXTERIOR MIRROR CONTROL OUTPUT CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: ~~DSM~~ [C341D](#) .
- Measure the resistance between LH exterior mirror [C521](#), harness side and ~~DSM~~ [C341D](#), harness side using the following chart:

DTC	LH Exterior Mirror	Circuit(s)	DSM
B1667	C521 Pin 3	CPM29 (GN)	C341D Pin 4
B1667	C521 Pin 4	CPM03 (BU/WH)/ CPM26 (WH)	C341D Pin 2
B1668	C521 Pin 5	CPM03 (BU/WH)/ CPM26 (WH)	C341D Pin 2
B1668	C521 Pin 6	CPM28 (BU/BN)	C341D Pin 5



N0092812

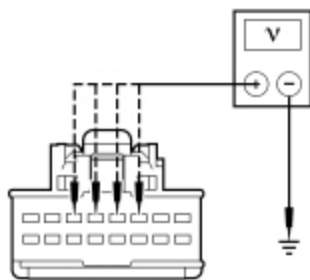
Are the resistances less than 3 ohms?

Yes	GO to D25 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D12 CHECK THE RH EXTERIOR MIRROR FOR VOLTAGE WHILE COMMANDING THE DSM PIDS

- Disconnect: RH Exterior Mirror [C615](#).
- Enter the following diagnostic mode on the scan tool: ~~DSM~~ DataLogger .
- **DTC B1669:** Measure the voltage between ground and RH exterior mirror:
 - [C615](#) Pin 3, circuit CPM34 (WH/BN), harness side while toggling the ~~DSM~~ PID Passenger's Mirror Up (PR_UP) to On.
 - [C615](#) Pin 4, circuit CPM31 (YE/GY), harness side while toggling the ~~DSM~~ PID Passenger's Mirror Down (PR_DOWN) to On.
- **DTC B1670:** Measure the voltage between ground and RH exterior mirror:

- [C615](#) Pin 5, circuit CPM31 (YE/GY), harness side while toggling the ~~DSM~~ PID Passenger's Mirror Left (PR_LEFT) to On.
- [C615](#) Pin 6, circuit CPM33 (WH/VT), harness side while toggling the ~~DSM~~ PID Passenger's Mirror Right (PR_RIGHT) to On.



N0092811

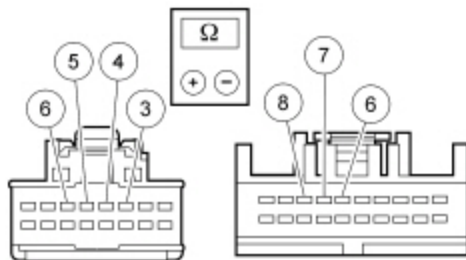
Do the voltages momentarily change from 0 volts to greater than 10 volts when the active command is toggled ON?

Yes	CHECK the RH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new RH exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to D13 .

D13 CHECK THE RH EXTERIOR MIRROR CONTROL OUTPUT CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: ~~DSM~~ [C341D](#) .
- Measure the resistance between RH exterior mirror [C615](#), harness side and ~~DSM~~ [C341D](#), harness side using the following chart:

DTC	RH Exterior Mirror	Circuit	DSM
B1669	C615 Pin 3	CPM34 (WH/BN)	C341D Pin 7
B1669	C615 Pin 4	CPM31 (YE/GY)	C341D Pin 8
B1670	C615 Pin 5	CPM31 (YE/GY)	C341D Pin 8
B1670	C615 Pin 6	CPM33 (WH/VT)	C341D Pin 6



N0092813

Are the resistances less than 3 ohms?

Yes	GO to D25 .
No	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D14 CHECK THE EXTERIOR MIRROR CONTROL SWITCH

- Disconnect: Exterior Mirror Control Switch C527 .
- Carry out the Exterior Mirror Control Switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

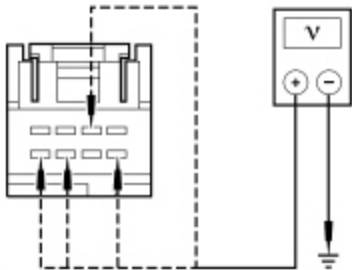
Did the exterior mirror control switch pass the component test?

Yes	GO to D15 .
No	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D15 CHECK THE MIRROR CONTROL CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~DSM~~ [C341D](#) .
- Ignition ON.
- Measure the voltage between exterior mirror control switch [C527](#), harness side and ground using the following chart:

DTC	Exterior Mirror Control Switch	Circuit
B1733	C527 Pin 5	CPM17 (BU/GN)
B1737	C527 Pin 8	CPM16 (BN/BU)
B1741	C527 Pin 2	CPM21 (YE/VT)
B1745	C527 Pin 7	CPM20 (BN/WH)



N0077237

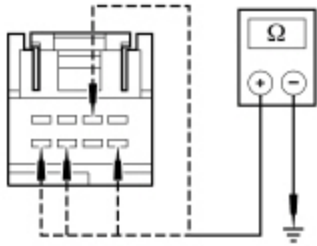
Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to D16 .

D16 CHECK THE EXTERIOR MIRROR SWITCH CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between exterior mirror control switch [C527](#), harness side and ground using the following chart:

DTC	Exterior Mirror Control Switch	Circuit
B1733	C527 Pin 5	CPM17 (BU/GN)
B1737	C527 Pin 8	CPM16 (BN/BU)
B1741	C527 Pin 2	CPM21 (YE/VT)
B1745	C527 Pin 7	CPM20 (BN/WH)



N0099045

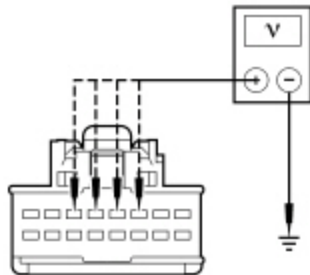
Is the resistance greater than 10,000 ohms?

Yes	GO to D25 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D17 CHECK THE LH EXTERIOR MIRROR CONTROL OUTPUT CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: LH Exterior Mirror [C521](#).
- Disconnect: DSM [C341D](#) .
- Ignition ON.
- Measure the voltage between LH exterior mirror, harness side and ground using the following chart:

LH Exterior Mirror	Circuit
C521 Pin 4	CPM03 (BU/WH)
C521 Pin 5	CPM03 (BU/WH)
C521 Pin 3	CPM29 (GN)
C521 Pin 6	CPM28 (BU/BN)



N0092811

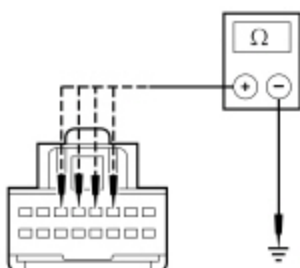
Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to D18 .

D18 CHECK THE LH EXTERIOR MIRROR CONTROL OUTPUT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between LH exterior mirror, harness side and ground using the following chart:

LH Exterior Mirror	Circuit
C521 Pin 4	CPM03 (BU/WH)
C521 Pin 5	CPM03 (BU/WH)
C521 Pin 3	CPM29 (GN)
C521 Pin 6	CPM28 (BU/BN)



N0121415

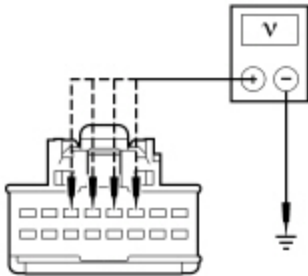
Are the resistances greater than 10,000 ohms?

Yes	CHECK the LH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for shorted circuits or damaged and pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D19 CHECK THE RH EXTERIOR MIRROR CONTROL OUTPUT CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: RH Exterior Mirror [C615](#).
- Disconnect: DSM [C341D](#) .
- Ignition ON.
- Measure the voltage between RH exterior mirror [C615](#), harness side and ground using the following chart:

RH Exterior Mirror	Circuit
C615 Pin 4	CPM31 (YE/GY)
C615 Pin 5	CPM31 (YE/GY)
C615 Pin 6	CPM33 (WH/VT)
C615 Pin 3	CPM34 (WH/BN)



N0092811

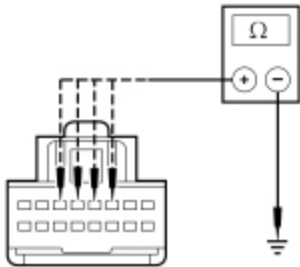
Is any voltage present?

Yes	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to D20 .

D20 CHECK THE RH EXTERIOR MIRROR CONTROL OUTPUT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between RH exterior mirror [C615](#), harness side and ground using the following chart:

RH Exterior Mirror	Circuit
C615 Pin 4	CPM31 (YE/GY)
C615 Pin 5	CPM31 (YE/GY)
C615 Pin 6	CPM33 (WH/VT)
C615 Pin 3	CPM34 (WH/BN)



N0121415

Are the resistances greater than 10,000 ohms?

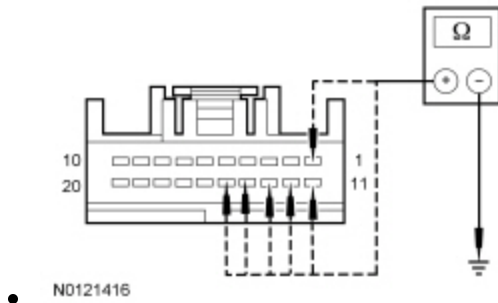
Yes	CHECK the RH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for shorted circuits or damaged and pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D21 CHECK THE MIRROR POSITION SENSOR CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: ~~DSM~~ [C341D](#) .
- Disconnect: **DTC B2312 or B2316:** RH Exterior Mirror [C615](#).
- Disconnect: **DTC B2320 or B2324:** LH Exterior Mirror [C521](#).
- Measure the resistance between ~~DSM~~ , harness side and ground using the following chart:

DSM	Circuit
C341D Pin 1	LPM30 (GY/VT)

DSM	Circuit
C341D Pin 11	RPM30 (YE)
C341D Pin 12	VPM36 (BN/YE)
C341D Pin 13	VPM35 (YE/BU)
C341D Pin 14	VPM38 (BN/GN)
C341D Pin 15	VPM37 (BU/OG)



Are the resistances greater than 10,000 ohms?

Yes	GO to D22 .
No	REPAIR the affected circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D22 CHECK THE MIRROR POSITION SENSOR CIRCUITS FOR AN OPEN

- Measure the resistance between DSM [C341D](#), harness side and exterior mirror [C521](#) (LH)/[C615](#) (RH), harness side using the following charts:

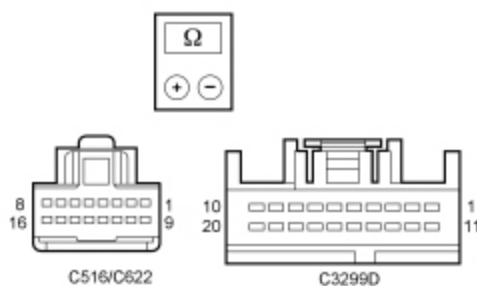
DTCs B2312 or B2316

RH Exterior Mirror	Circuit	DSM
C615 Pin 12	LPM30 (GY/VT)	C341D Pin 1
C615 Pin 15	RPM30 (YE)	C341D Pin 11
C615 Pin 14	VPM38 (BN/GN)	C341D Pin 14
C615 Pin 13	VPM37 (BU/OG)	C341D Pin 15

DTCs B2320 or B2324

DSM	Circuit	LH Exterior Mirror
C341D Pin 1	LPM30 (GY/VT)	C521 Pin 12

DSM	Circuit	LH Exterior Mirror
C341D Pin 11	RPM30 (YE)	C521 Pin 15
C341D Pin 12	VPM36 (BN/YE)	C521 Pin 14
C341D Pin 13	VPM35 (GY/VT)	C521 Pin 13



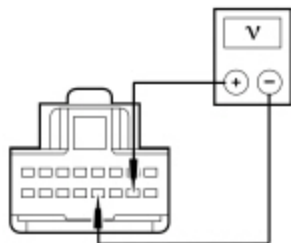
N0120818

Are the resistances less than 3 ohms?

Yes	GO to D23 .
No	REPAIR the affected circuit. CLEAR the DTCs. REPEAT the self test. TEST the system for normal operation.

D23 CHECK THE MIRROR POSITION SENSOR FEED AND RETURN FROM THE DSM

- Connect: ~~DSM~~ [C341D](#) .
- Ignition ON.
- For DTCs **B2312** or **B2316**, measure the voltage between RH exterior mirror [C615](#) Pin 12, circuit LPM30 (GY/VT), harness side and RH exterior mirror [C615](#) Pin 15, circuit RPM30 (YE), harness side.
- For DTCs **B2320** or **B2324**, measure the voltage between LH exterior mirror [C521](#) Pin 12, circuit LPM30 (GY/VT), harness side and LH exterior mirror [C521](#) Pin 15, circuit RPM30 (YE), harness side.



N0120819

Is the voltage approximately 5 volts?

Yes	For DTCs B2312 or B2316, CHECK the RH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for open or shorted circuits and damaged or pushed-out pins. For DTCs B2320 or B2324, CHECK the LH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for open or shorted circuits and damaged or pushed-out pins.
-----	--

If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

No GO to [D25](#).

D24 CHECK THE EXTERIOR MIRROR POSITION SENSOR CIRCUITS FOR A SHORT TO VOLTAGE

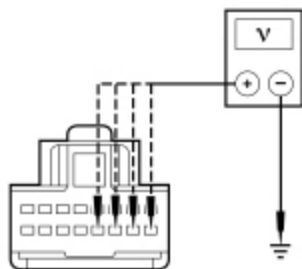
- Ignition OFF.
- Disconnect: ~~DSM~~ [C341D](#) .
- Disconnect: **DTC B2314 or B2318:** RH Exterior Mirror [C615](#).
- Disconnect: **DTC B2322 or B2326:** LH Exterior Mirror [C521](#).
- Ignition ON.
- Measure the voltage between exterior mirror [C521](#) (LH)/ [C615](#) (RH), harness side and ground using the following charts:

DTCs B2314 or B2318

RH Exterior Mirror	Circuit
C615 Pin 12	LPM30 (GY/VT)
C615 Pin 15	RPM30 (YE)
C615 Pin 14	VPM38 (BN/GN)
C615 Pin 13	VPM37 (BU/OG)

DTCs B2322 or B2326

LH Exterior Mirror	Circuit
C521 Pin 12	LPM30 (GY/VT)
C521 Pin 15	RPM30 (YE)
C521 Pin 14	VPM36 (BN/YE)
C521 Pin 13	VPM35 (GY/VT)



N0120820

Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	For DTCs B2314 or B2318, CHECK the RH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for open or shorted circuits and damaged or pushed-out pins. For DTCs B2322 or B2326, CHECK the LH exterior mirror jumper harness between the vehicle harness and the exterior mirror motors for open or shorted circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. If the jumper harness is OK, INSTALL a new exterior mirror motor. REFER to Exterior Mirror Motor in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D25 CHECK THE DSM FOR CORRECT OPERATION

- Disconnect all DSM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new DSM . REFER to Section 501-10. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test E: The Power Folding Mirrors Do Not Operate/Do Not Operate Correctly (Without Power Telescoping Mirrors)

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

NOTE: The power folding mirrors must be synchronized anytime the RH, LH or both power folding mirrors are folded or unfolded without using the power folding mirror control switch, or if a new power folding mirror has been installed. Refer to Exterior Mirror Motor Synchronization — Power Folding in this section.

NOTE: If the power folding mirrors are folded or unfolded continuously (6 to 10 cycles), the power lockout feature will disable the system for approximately 3-10 minutes to prevent damage to the power fold motors. After 3-10 minutes have elapsed, normal operation resumes.

NOTE: Verify that the power folding mirrors are not binding or obstructed.

Normal Operation

When the exterior mirror control switch is set to the center position and pressed rearward to the FOLD/UNFOLD position, the exterior mirror control switch sends a signal to the DSM which then supplies power and ground to the exterior mirrors to fold or unfold the mirrors.

- DTC B112B (Mirror Power Fold Switch Input Short to Battery) — Short to battery on mirror power fold switch input circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Exterior mirror control switch
- Exterior mirror
- Driver Seat Module (DSM)

PINPOINT TEST E : THE POWER FOLDING MIRRORS DO NOT OPERATE/DO NOT OPERATE CORRECTLY (WITHOUT POWER TELESCOPING MIRRORS)

NOTE: Before running a self-test diagnostic or retrieving DTCs, the power fold switch must be inactive for at least 30 seconds. Retrieving DTCs before 30 seconds results in false DTCs.

E1 CHECK FOR DTCS FROM THE DSM

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — DSM .

Are any DTCs retrieved?

Yes	If DTC B112B is present, GO to E8 . For all other DSM DTCs, REFER to Section 501-10.
No	GO to E2 .

E2 CHECK THE OPERATION OF THE POWER FOLDING MIRRORS

- Position the exterior mirror control switch to the center position.
- Press the exterior mirror control switch rearward towards the FOLD/UNFOLD position to activate the power folding mirrors.

Do the RH and LH power folding mirror operate?

Yes	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector or the power folding mirror power lockout feature may have been active. TEST the system for normal operation.
-----	---

No	If both the LH and RH power folding mirrors are inoperative, GO to E3 . If only the LH power folding mirror is inoperative, GO to E10 . If only the RH power folding mirror is inoperative, GO to E11 .
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E3 CHECK THE EXTERIOR MIRROR CONTROL SWITCH

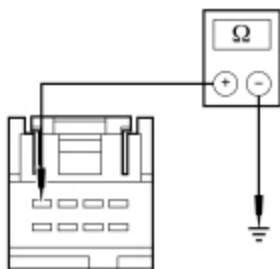
- Ignition OFF.
- Disconnect: Exterior Mirror Control Switch [C527](#).
- Carry out the Exterior Mirror Control Switch component test. Refer to Wiring Diagrams Cell [149](#) for component testing.

Did the exterior mirror control switch pass the component test?

Yes	GO to E4 .
No	INSTALL a new exterior mirror control switch. REFER to Exterior Mirror Control Switch — Aero in this section. TEST the system for normal operation.

E4 CHECK THE FOLDING MOTOR CONTROL CIRCUIT FOR A SHORT TO GROUND

- Disconnect: DSM [C341C](#) .
- Measure the resistance between exterior mirror control switch [C527](#) Pin 4, circuit CPM22 (GY/BN) harness side and ground.



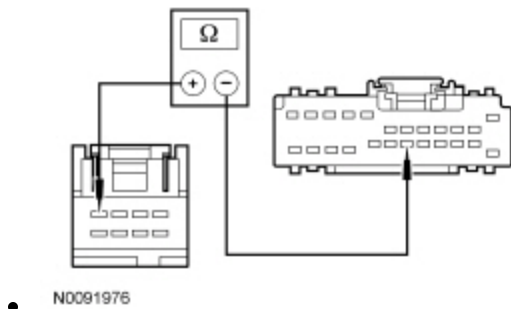
N0065191

Is the resistance greater than 10,000 ohms?

Yes	GO to E5 .
No	REPAIR the circuit. TEST the system for normal operation.

E5 CHECK THE FOLDING MOTOR CONTROL CIRCUIT FOR AN OPEN

- Measure the resistance between exterior mirror control switch [C527](#) Pin 4, circuit CPM22 (GY/BN), harness side and [C341C](#) Pin 18, circuit CPM22 (GY/BN), harness side.

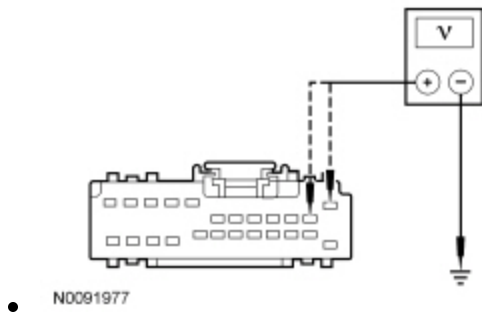


Is the resistance less than 3 ohms?

Yes	GO to E6 .
No	REPAIR the circuit. SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.

E6 CHECK THE FOLDING MIRROR OUTPUT CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: LH Exterior Mirror [C521](#) and RH Exterior Mirror [C615](#).
- Ignition ON.
- Measure the voltage between ground and DSM :
 - [C341C](#) Pin 2, circuit CPM04 (VT/BN), harness side.
 - [C341C](#) Pin 1, circuit CPM39 (GN/VT), harness side.

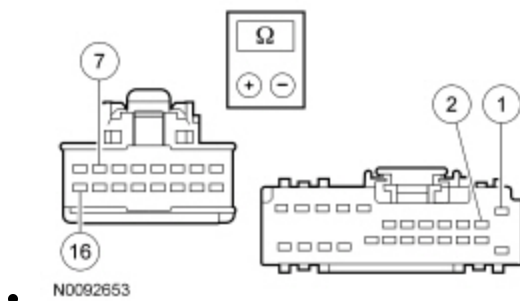


Is any voltage present?

Yes	REPAIR the affected circuit(s). SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.
No	GO to E7 .

E7 CHECK THE FOLDING MIRROR OUTPUT CIRCUITS FOR AN OPEN

- Ignition OFF.
- Measure the resistance between exterior mirror:
 - [C521](#) Pin 7, circuit CPM04 (VT/BN), harness side and [C341C](#) Pin 2, circuit CPM04 (VT/BN), harness side.
 - [C521](#) Pin 16, circuit CPM39 (GN/VT), harness side and [C341C](#) Pin 1, circuit CPM39 (GN/VT), harness side.

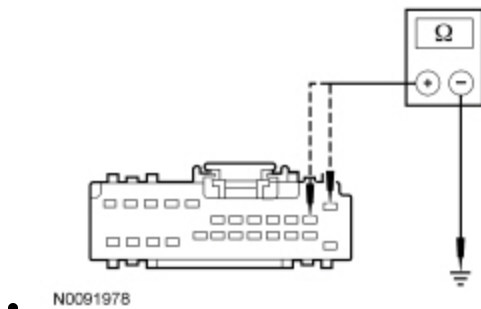


Are the resistances less than 3 ohms?

Yes	GO to E8 .
No	REPAIR the circuit(s). SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.

E8 CHECK THE FOLDING MIRROR OUTPUT CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between ground and DSM :
 - [C341C](#) Pin 2, circuit CPM04 (VT/BN), harness side.
 - [C341C](#) Pin 1, circuit CPM39 (GN/VT), harness side.

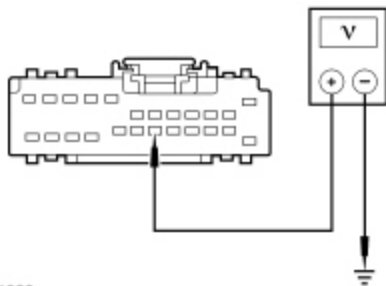


Are the resistances greater than 10,000 ohms?

Yes	GO to E9 .
No	REPAIR the circuit(s). SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.

E9 CHECK THE FOLDING MOTOR CONTROL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Exterior Mirror Control Switch [C527](#).
- Disconnect: DSM [C341C](#) .
- Ignition ON.
- Measure the voltage between DSM [C341C](#) Pin 18, circuit CPM22 (GY/BN), harness side and ground.



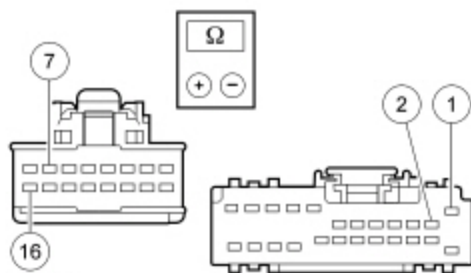
N0091980

Is any voltage present ?

Yes	REPAIR the circuit. SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to E12 .

E10 CHECK THE LH POWER FOLD EXTERIOR MIRROR CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Exterior Mirror [C521](#).
- Disconnect: ~~DSM~~ [C341C](#) .
- Measure the resistance between exterior mirror:
 - [C521](#) Pin 7, circuit CPM04 (VT/BN), harness side and ~~DSM~~ [C341C](#) Pin 2, circuit CPM04 (VT/BN), harness side.
 - [C521](#) Pin 16, circuit CPM39 (GN/VT), harness side and ~~DSM~~ [C341C](#) Pin 1, circuit CPM39 (GN/VT), harness side.



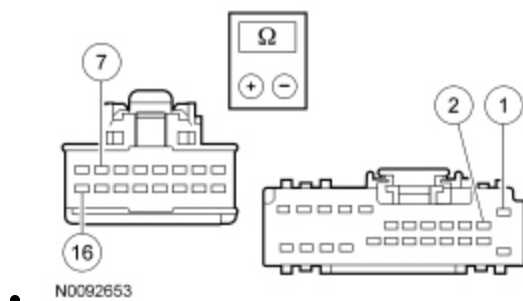
N0092653

Are the resistances less than 3 ohms?

Yes	CHECK the LH exterior mirror jumper harness between the vehicle harness and the exterior mirror glass for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new LH exterior mirror. REFER to Exterior Mirror in this section. SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.

E11 CHECK THE RH POWER FOLD EXTERIOR MIRROR CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: RH Exterior Mirror [C615](#).
- Disconnect: DSM [C341C](#) .
- Measure the resistance between exterior mirror:
 - [C615](#) Pin 7, circuit CPM04 (VT/BN), harness side and DSM [C341C](#) Pin 2, circuit CPM04 (VT/BN), harness side.
 - [C615](#) Pin 16, circuit CPM39 (GN/VT), harness side and DSM [C341C](#) Pin 1, circuit CPM39 (GN/VT), harness side.



Are the resistances less than 3 ohms?

Yes	CHECK the RH exterior mirror jumper harness between the vehicle harness and the exterior mirror glass for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new RH exterior mirror. REFER to Exterior Mirror in this section. SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. TEST the system for normal operation.

E12 CHECK THE DSM FOR CORRECT OPERATION

- Disconnect all DSM connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all DSM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new DSM . REFER to Section 419-10. CLEAR the DTCs. SYNCHRONIZE the power folding mirrors. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section. REPEAT the self-test. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector or the power folding mirror power lockout feature may have been active.

	CLEAR the DTCs. REPEAT the self-test.
--	---------------------------------------

Pinpoint Test F: The Heated Exterior Mirror is Inoperative — One or Both Mirrors

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

Refer to Wiring Diagrams Cell [56](#) , Heated Window for schematic and connector information.

Normal Operation

The heated mirror/rear window defrost function can be activated at one of the three following locations: HVAC module, FCIM or 8- inch (203mm) touch screen. When the engine is running and the heated mirror/rear window defrost system is activated, the HVAC module activates the rear window defrost relay located in the BJB . When the rear window defrost relay is active, voltage is supplied to the LH and RH exterior mirror glass and spotter mirror glass (if equipped). If the sliding glass is opened while the defrost system is active, the heated mirrors continue to operate and the rear window defrost switch LED is no longer illuminated. The LH and RH mirrors do not share the same ground circuits.

- DTC B1C83:11 (Rear Defog Relay: Circuit Short to Ground) — Sets when the HVAC module senses a short to ground on the heated mirror/rear window defrost relay control circuit.
- DTC B1C83:12 (Rear Defog Relay: Circuit Short to Battery) — Sets when the HVAC module senses a short to voltage on the heated mirror/rear window defrost relay control circuit (manual climate control only).
- DTC B1C83:13 (Rear Defog Relay: Circuit Open) — Sets when the HVAC module senses an open on the heated mirror/rear window defrost relay control circuit.

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Rear window defrost relay
- Wiring, terminals or connectors
- Exterior mirror glass
- Exterior spotter mirror glass
- Exterior mirror
- HVAC module
- FCIM

PINPOINT TEST F : THE HEATED EXTERIOR MIRROR IS INOPERATIVE — ONE OR BOTH MIRRORS

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

F1 RETRIEVE DTCS FROM THE HVAC MODULE
--

- | |
|--|
| <ul style="list-style-type: none">• Ignition ON.• Enter the following diagnostic mode on the scan tool: Self Test HVAC Module . |
|--|

Is DTC B1C83:11 retrieved?

Yes	GO to F2 .
No	GO to F4 .

F2 PERFORM THE HVAC MODULE SELF TEST WITH A KNOWN GOOD RELAY INSTALLED

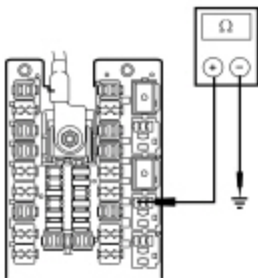
- Ignition OFF.
- Replace the rear window defrost relay with a known good relay.
- Ignition ON.
- Using a scan tool, clear the HVAC module DTCs.
- Perform the HVAC module self test.

Is DTC B1C83:11 present?

Yes	GO to F3 .
No	INSTALL a new rear window defrost relay. TEST the system for normal operation.

F3 CHECK THE REAR WINDOW DEFROST RELAY CONTROL CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect rear window defrost relay.
- Disconnect HVAC module Dual Automatic Temperature Control (DATC) or Single Automatic Temperature Control [C228](#) Pin 25 or Electronic Manual Temperature Control (EMTC) [C294A](#) Pin 25.
- Measure the resistance between: ~~BJB~~ rear window defrost relay pin 2 and ground.



N0148758

Is the resistance greater than 10,000 ohms?

Yes	GO to F15 .
No	REPAIR the circuit. TEST the system for normal operation.

F4 CHECK THE OPERATION OF THE HEATED EXTERIOR MIRRORS GLASS

NOTE: The defrost system activates for 10 minutes.

- START the engine.
- Activate the heated mirror/rear window defrost system.
- Using a infrared temperature gun measure the temperature of the exterior mirrors glass for approximately 30 seconds noting changes in temperature.

Does at least one of the heated exterior mirrors rise in temperature?

Yes	GO to F5 .
No	GO to F7 .

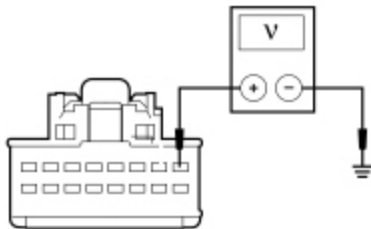
F5 CHECK THE VOLTAGE TO THE INOPERATIVE HEATED EXTERIOR REAR VIEW MIRROR GLASS

- Ignition OFF.
- Disconnect: The Inoperative Heated Exterior Rear View Mirror [C521](#) (LH exterior mirror) or [C615](#) (RH exterior mirror) .
- Start the engine.
- **NOTE:** *The defrost system activates for 10 minutes.*

Activate the heated mirror/rear window defrost system.

- Measure the voltage between the inoperative heated exterior rear view mirror, harness side and ground as follows:

Connector-Pin	Circuit
C521 Pin 1 (LH exterior mirror)	CRD17 (GY)
C615 Pin 1 (RH exterior mirror)	CRD17 (GY)



N0121931

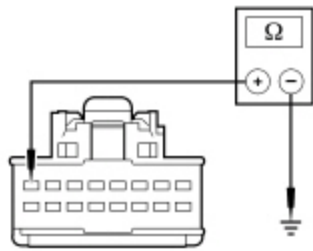
Is the voltage greater than 11 volts?

Yes	GO to F6 .
No	REPAIR the circuit for an open, TEST the system for normal operation.

F6 CHECK THE INOPERATIVE EXTERIOR REAR VIEW MIRROR GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the inoperative heated exterior rear view mirror, harness side and ground as follows:

Connector-Pin	Circuit
C521 Pin 8 (LH exterior mirror)	GD134 (BK/VT)
C615 Pin 8 (RH exterior mirror)	GD139 (BK/YE)



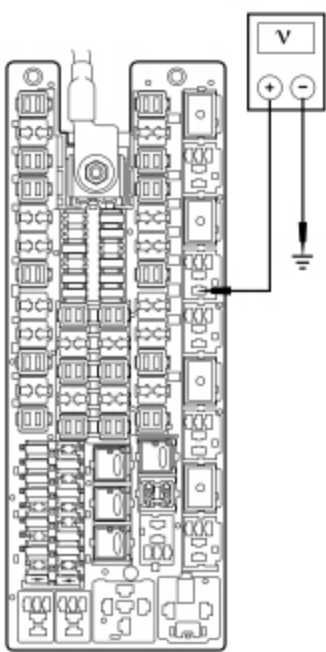
• N0103955

Is the resistance less than 3 ohms?

Yes	CHECK the inoperative rear view mirror exterior jumper harness between the vehicle harness and the exterior mirror glass or exterior mirror spotter mirror glass for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness cannot be repaired, INSTALL a new exterior mirror. REFER to Exterior Mirror in this section. TEST the system for normal operation. If the jumper harness is OK, INSTALL a new exterior mirror glass. REFER to Exterior Mirror Glass — Aero or Exterior Mirror Glass — Trailer Tow or Exterior Mirror Glass — Spotter, Trailer Tow in this section. TEST the system for normal operation.
No	REPAIR the circuit in question for an open. TEST the system for normal operation.

F7 CHECK THE REAR WINDOW DEFROST RELAY CONTROL CIRCUIT FOR SUPPLY VOLTAGE

- Ignition OFF.
- Disconnect: ~~HYAC~~ Module Dual Automatic Temperature Control (DATC) or Single Automatic Temperature Control [C228](#) Pin 25 or Electronic Manual Temperature Control (EMTC) [C294A](#) Pin 25.
- Disconnect: Rear Window Defrost Relay .
- Measure the voltage between ~~BJB~~ rear window defrost relay pin 3, circuit SBB32 (VT/RD), harness side and ground.



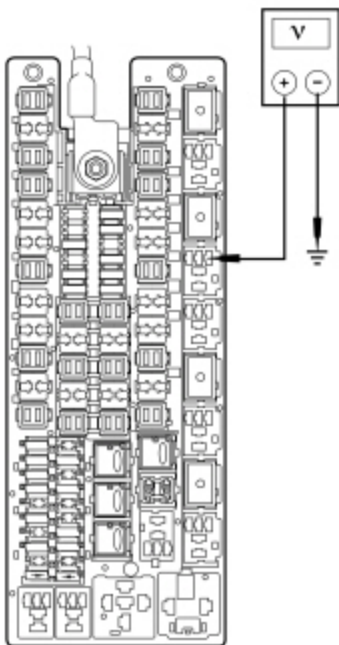
N0148562

Is the voltage greater than 11 volts?

Yes	GO to F8 .
No	Verify that BJB fuse 32 (40A) is OK. If OK, REPAIR circuit SBB32 for an open. If not OK, REFER to the wiring diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

F8 CHECK THE REAR WINDOW DEFROST RELAY CONTROL VOLTAGE SUPPLY

- Ignition ON.
- Measure the voltage between the Battery Junction Box (BJB) rear window defrost relay pin 1, circuit CBB54 (VT/OG) and ground. TEST the system for normal operation.



N0123457

Is the voltage greater than 11 volts?

Yes	GO to F9 .
No	VERIFY that BJB fuse 54 (5A) is OK. If OK, REPAIR the circuit CBB54 (VT/OG) for an open. If not OK, REFER to the Wiring Diagrams manual to identify possible causes of the circuit short.

F9 CHECK THE REAR WINDOW DEFROST RELAY

- Ignition OFF.
- INSTALL a known good relay for the rear window defrost relay.
- START the engine.
- **NOTE:** *The defrost system activates for 10 minutes.*

Activate the heated mirror/rear window defrost switch to the ON position.

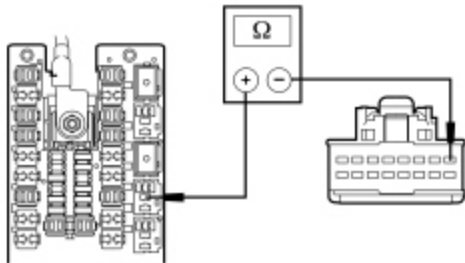
- Using a infrared temperature gun measure the temperature of the exterior mirrors glass for approximately 30 seconds noting changes in temperature.

Does the temp of the exterior mirrors glass rise?

Yes	INSTALL a new rear window defrost relay. TEST the system for normal operation.
No	GO to F10 .

F10 CHECK THE HEATED EXTERIOR MIRROR SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Rear Window Defrost Relay .
- Disconnect: LH Exterior Mirror .
- Measure the resistance between: BJB rear window defrost relay pin 5, circuit CRD06 (BN/YE) and [C521](#) Pin 1, circuit CRD17 (GY) harness side.



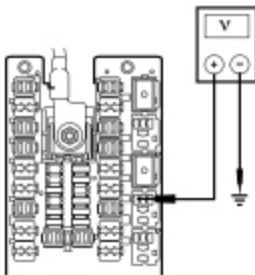
Is the resistance less than 3 ohms?

Yes	GO to F11 .
No	VERIFY fuse 56 (15A) is OK. If OK, REPAIR the circuit for an open. If not OK, REFER to the Wiring Diagram Manual to identify the possible causes for the circuit short. TEST the system for

normal operation.

F11 CHECK THE REAR WINDOW DEFROST RELAY CONTROL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: HVAC Module Dual Automatic Temperature Control (DATC) or Single Automatic Temperature Control [C228](#) Pin 25 or Electronic Manual Temperature Control (EMTC) [C294A](#) Pin 25.
- Ignition ON.
- Measure the voltage between Battery Junction Box (BJB) rear window defrost relay pin 2, circuit CRD02 (BN/BU) and ground.



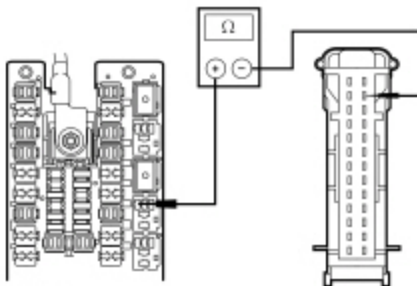
N0148568

Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to F12 .

F12 CHECK THE REAR WINDOW DEFROST RELAY CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the Battery Junction Box (BJB) rear window defrost relay pin 2, circuit CRD02 (BN/BU) and the HVAC Dual Automatic Temperature Control (DATC) or Single Automatic Temperature Control [C228](#) Pin 25 or Electronic Manual Temperature Control (EMTC) [C294A](#) Pin 25.



N0148569

Is the resistance less than 3 ohms?

Yes	If equipped with the 4.2- inch (107 mm) or 8- inch (203 mm) display, GO to F13 . If not equipped with the 4.2- inch (107 mm) or 8- inch (203 mm) display, GO to F15 .
No	REPAIR the circuit. TEST the system for normal operation.

F13 CHECK THE REAR WINDOW DEFROST SWITCH PID

- Connect: Connect The Rear Window Defrost Relay .
- Connect: Connect HVAC Module .
- Ignition ON.
- Using a scan tool, view the FCIM PIDs.
- While observing the (CC_SW_R_DEF) PID, press and release the heated mirror/rear window defrost switch.

Does the PID agree with the rear window defrost switch state?

Yes	GO to F14 .
No	INSTALL a new FCIM , Refer to the appropriate section in Group 415 for the procedure.

F14 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, perform the network test.

Do the HVAC module and the FCIM pass the network tests?

Yes	GO to F15 .
No	REFER to Section 418-00.

F15 CHECK FOR CORRECT HVAC MODULE OPERATION

- Disconnect: All HVAC Module Electrical Connectors .
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect: All HVAC Module Electrical Connectors And Make Sure They Seat Correctly .
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test G: DTCs B106E and B106F

Refer to Wiring Diagrams Cell [124](#) , Power Mirrors for schematic and connector information.

Normal Operation

The Driver Seat Module (DSM) controls the output of the exterior mirrors using solid state drivers. A DTC sets when an overload occurs on any of these drivers. The module also tracks the number of repetitive faults on each of these circuits. The module compares this number of overloads to 3 progressive thresholds established for each circuit. If the 3 thresholds have not been exceeded, then the DTC of the affected circuit can be cleared by eliminating the fault, then clearing the DTCs and running the self-test. At the point that each of the first 2 thresholds are exceeded, DTC B106E sets along with the DTC relating to the affected circuit. Once the final threshold has been exceeded, the affected output is permanently disabled and DTC B106F sets at which time the DSM must be replaced.

- DTC B106E (Solid State Driver Disabled Due to Short Circuit) — sets when the DSM has disabled a circuit due to a repetitive fault causing a circuit overload. A corresponding DTC for the circuit in question also sets.
- DTC B106F (Module Disabled Due to External Fault) — sets when one or more output functions are permanently disabled due to a repetitive circuit overload fault. When DTC B106F is present, a new DSM must be installed after the fault condition has been corrected.

This pinpoint test is intended to diagnose the following:

- Output circuit short to ground or voltage
- DSM

PINPOINT TEST G : DTCS B106E AND B106F	
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
NOTICE: If DTC B106F is present, correct the fault condition prior to installing a new DSM . Failure to correct the fault condition first may cause damage to the new DSM , resulting in a repeat repair.	
G1 REVIEW THE DSM DTCS FROM THE SELF-TEST	
• Check the recorded results from the DSM self-test.	
Is DTC B106F present?	
Yes	CORRECT the cause of any other DTCs present first. When other DTCs are corrected, INSTALL a new DSM . For additional information, REFER to Section 501-10. TEST the system for normal operation.
No	GO to G2 .
G2 REPAIR OTHER DTCS	
• Repair the cause of any DTCs present, other than DTC B106E. Refer to the Driver Seat Module (DSM) DTC Chart in this section. • Clear the DTCs. • Carry out the DSM self-test.	
Is DTC B106E still present?	
Yes	The fault causing the output state to be disabled is still present. REPAIR the cause of the concern.

	CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. The concern was caused by a fault in one of the output circuits.

Pinpoint Test I: The Power Telescoping Mirrors Do Not Operate/Do Not Operate Correctly

Refer to Wiring Diagrams Cell 124, Power Mirrors for schematic and connector information.

Normal Operation

The power telescoping mirror control switch controls movement by reversing the polarity of the voltage and ground it supplies to the power telescoping mirror motors. The RH and LH power telescoping mirrors extend and retract at the same time and cannot be controlled independently.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Power telescoping mirror control switch
- Power telescoping mirror

PINPOINT TEST I : THE POWER TELESCOPING MIRRORS DO NOT OPERATE/DO NOT OPERATE CORRECTLY

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

I1 CHECK POWER TELESCOPING MIRROR OPERATION

- Ignition ON.
- Press the power telescoping mirror control switch to activate the power telescoping mirrors inward and outward.

Are both power telescoping mirrors inoperative?

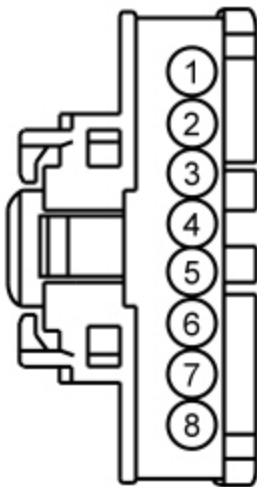
Yes	GO to I2 .
No	For the LH mirror GO to I4 . For the RH mirror GO to I8 .

I2 CHECK FOR VOLTAGE TO THE POWER TELESCOPING MIRROR CONTROL SWITCH

- Ignition OFF.
- Disconnect: Power Telescoping Mirror Control Switch [C577](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

C577 Pin 4	CBP32 (GN/VT)	-	Ground
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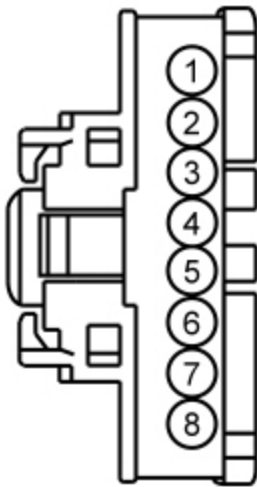
Is the voltage greater than 11 volts?

Yes	GO to I3 .
No	VERIFY the BCM fuse 32 (15A) is OK. If OK, REPAIR the circuit for an open. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

I3 CHECK FOR GROUND TO THE POWER TELESCOPING MIRROR CONTROL SWITCH

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 5	GD134 (BK/VT)	-	Ground



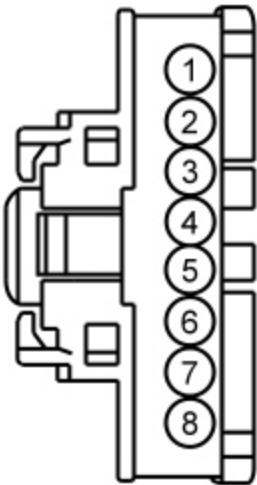
Is the resistance less than 3 ohms?

Yes	INSTALL a new power telescoping mirror control switch. REFER to Exterior Mirror Control Switch — Trailer Tow in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

I4 CHECK THE LH POWER TELESCOPING MIRROR CONTROL CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: LH Exterior Mirror [C521](#).
- Disconnect: Power Telescoping Mirror Control Switch [C577](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 3	CPM42 (YE/OG)	-	Ground
C577 Pin 7	CPM41 (VT)	-	Ground



Is any voltage present?

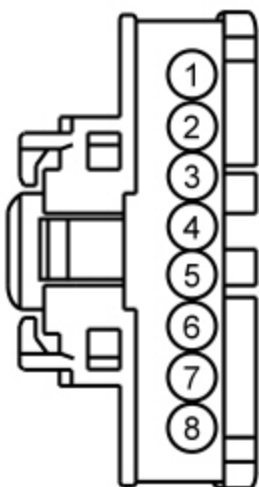
Yes	REPAIR the circuit(s). TEST the system for normal operation.
No	GO to I5 .

I5 CHECK THE LH POWER TELESCOPING MIRROR CONTROL CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 3	CPM42 (YE/OG)	-	Ground
C577 Pin 7	CPM41 (VT)	-	Ground



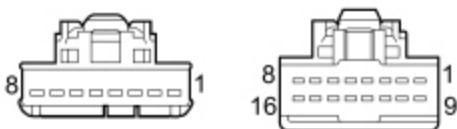
Are the resistances greater than 10,000 ohms?

Yes	GO to I6 .
No	REPAIR the circuit(s). TEST the system for normal operation.

I6 CHECK THE LH POWER TELESCOPING MIRROR CONTROL CIRCUITS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 3	CPM42 (YE/OG)	C521 Pin 2	CPM42 (YE/OG)
C577 Pin 7	CPM41 (VT)	C521 Pin 9	CPM41 (VT)



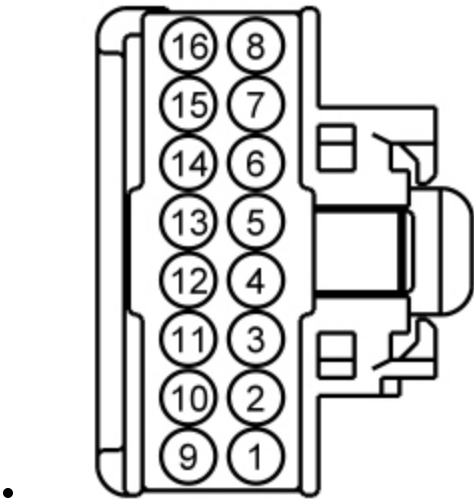
Are the resistances less than 3 ohms?

Yes	GO to 17 .
No	REPAIR the circuit(s). TEST the system for normal operation.

I7 CHECK THE VOLTAGE AVAILABLE TO THE TELESCOPING MIRROR MOTOR CIRCUIT AT THE LH SIDE EXTERIOR MIRROR

- Connect: Power Telescoping Mirror Control Switch [C577](#).
- Ignition ON.
- Press the power telescoping mirror control switch and measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C521 Pin 2	CPM42 (YE/OG)	-	Ground
C521 Pin 9	CPM41 (VT)	-	Ground



Is the voltage greater than 11 volts?

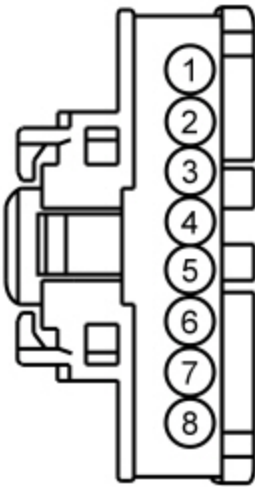
Yes	CHECK the LH exterior mirror jumper harness for open or shorted circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness is OK or cannot be repaired, INSTALL a new LH exterior mirror. REFER to Exterior Mirror in this section. TEST the system for normal operation.
No	INSTALL a new power telescoping mirror control switch. REFER to Exterior Mirror Control Switch — Trailer Tow in this section. TEST the system for normal operation.

I8 CHECK THE RH POWER TELESCOPING MIRROR CONTROL CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.

- Disconnect: RH Exterior Mirror [C615](#).
- Disconnect: Power Telescoping Mirror Control Switch [C577](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 2	CPM44 (GN/OG)	-	Ground
C577 Pin 6	CPM43 (WH/VT)	-	Ground



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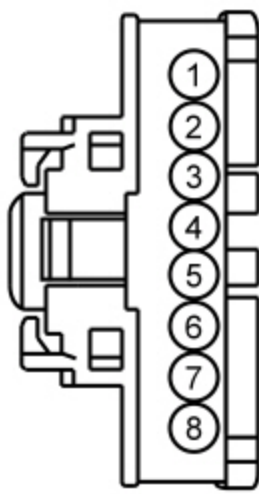
Is any voltage present?

Yes	REPAIR the circuit(s). TEST the system for normal operation.
No	GO to 19 .

19 CHECK THE RH POWER TELESOPING MIRROR CONTROL CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 2	CPM44 (GN/OG)	-	Ground
C577 Pin 6	CPM43 (WH/VT)	-	Ground



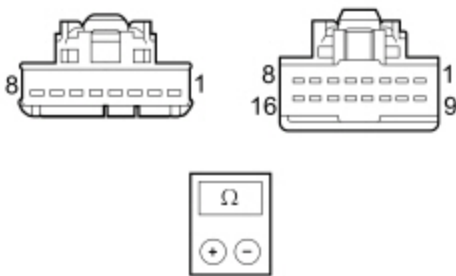
Are the resistances greater than 10,000 ohms?

Yes	GO to I10 .
No	REPAIR the circuit(s). TEST the system for normal operation.

I10 CHECK THE RH POWER TELESCOPING MIRROR CONTROL CIRCUITS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C577 Pin 2	CPM44 (GN/OG)	C615 Pin 2	CPM44 (GN/OG)
C577 Pin 6	CPM43 (WH/VT)	C615 Pin 9	CPM43 (WH/VT)



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Are the resistances less than 3 ohms?

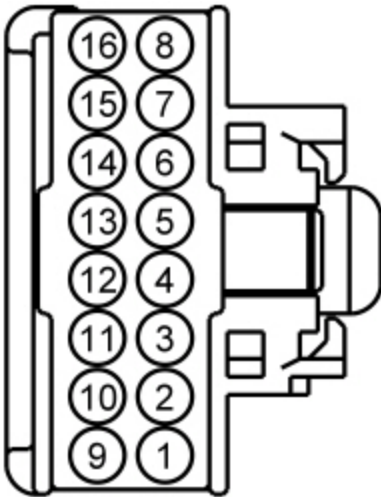
Yes	GO to I11 .
No	REPAIR the circuit(s). TEST the system for normal operation.

I11 CHECK THE VOLTAGE AVAILABLE TO THE TELESCOPING MIRROR MOTOR CIRCUIT AT THE RH SIDE EXTERIOR MIRROR

- Connect: Power Telescoping Mirror Control Switch [C577](#).

- Ignition ON.
- Press the power telescoping mirror control switch and measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C615 Pin 2	CPM44 (GN/OG)	-	Ground
C615 Pin 9	CPM43 (WH/VT)	-	Ground



Is the voltage greater than 11 volts?

Yes	CHECK the RH exterior mirror jumper harness for open or shorted circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness is OK or cannot be repaired, INSTALL a new RH exterior mirror. REFER to Exterior Mirror in this section. TEST the system for normal operation.
No	INSTALL a new power telescoping mirror control switch. REFER to Exterior Mirror Control Switch — Trailer Tow in this section. TEST the system for normal operation.

Pinpoint Test J: The Power Folding Mirrors Do Not Operate/Do Not Operate Correctly (With Power Telescoping Mirrors)

NOTE: The power folding mirrors may need to be synchronized when the RH, LH or both power folding mirrors are partially folded in toward the door manually, and then folded inward using the power folding mirror control switch. REFER to Exterior Mirror Motor Synchronization — Power Folding in this section.

NOTE: If the power folding mirrors are folded or unfolded continuously for 10 cycles, the power mirror fold relay will disable the system for 3-1/2 minutes to prevent damage to the power folding mirror motors. After 3-1/2 minutes have elapsed, normal operation will resume.

NOTE: Verify that the power folding mirrors are not binding or obstructed.

Refer to Wiring Diagrams Cell 124, Power Mirrors for schematic and connector information.

Normal Operation

When the power mirror fold relay receives a ground signal from the exterior mirror control switch, the power mirror fold relay will activate the power folding mirrors in the opposite direction of the last activation for approximately 20 seconds. The power mirror fold relay controls movement by reversing the polarity of the voltage and ground it supplies to the power folding mirror motors.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Exterior mirror control switch
- Power mirror fold relay
- Power folding mirror

PINPOINT TEST J : THE POWER FOLDING MIRRORS DO NOT OPERATE/DO NOT OPERATE CORRECTLY (WITH POWER TELESCOPING MIRRORS)

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

J1 CHECK POWER FOLDING MIRROR OPERATION

NOTE: Set the exterior mirror control switch to the center position.

- Ignition ON.
- Press the exterior mirror control switch rearward towards the FOLD/UNFOLD position to activate the power folding mirrors.

Are both power folding mirrors inoperative?

Yes	GO to J2 .
No	GO to J8 .

J2 CHECK THE EXTERIOR MIRROR CONTROL SWITCH

- Ignition OFF.
- Disconnect: Exterior Mirror Control Switch [C527](#).
- Carry out the Exterior Mirror Control Switch component test. Refer to Wiring Diagrams for component testing Refer to Wiring Diagrams Cell [149](#) for component testing.

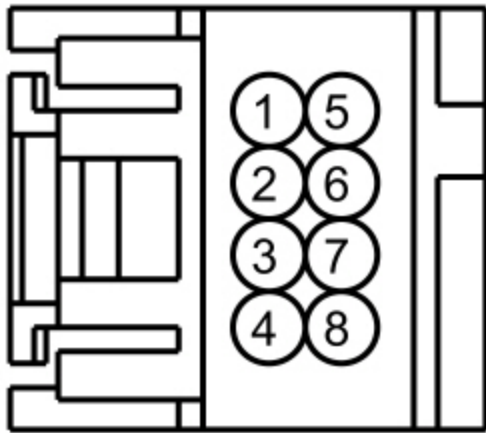
Is the exterior mirror control switch OK?

Yes	GO to J3 .
No	INSTALL a new exterior mirror control switch. Exterior Mirror Control Switch — Trailer Tow.

J3 CHECK THE FOLDING MIRROR CONTROL CIRCUIT FOR A SHORT TO GROUND

- Disconnect: Power Mirror Fold Relay [C3276](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C527 Pin 4	CPM22 (GY/BN)	-	Ground



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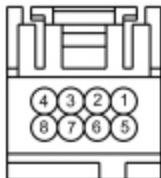
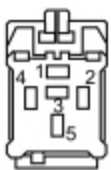
Is the resistance greater than 10,000 ohms?

Yes	GO to J4 .
No	REPAIR the circuit. TEST the system for normal operation.

J4 CHECK THE FOLDING MIRROR CONTROL CIRCUIT FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C527 Pin 4	CPM22 (GY/BN)	C3276 Pin 4	CPM22 (GY/BN)



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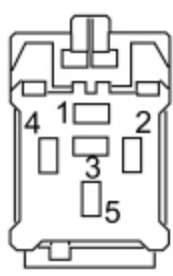
Is the resistance less than 3 ohms?

Yes	GO to J5 .
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No	REPAIR the circuit. TEST the system for normal operation.
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J5 CHECK FOR VOLTAGE TO THE POWER MIRROR FOLD RELAY

- Ignition ON.
-



- N0149489
- Measure the **voltage** between:

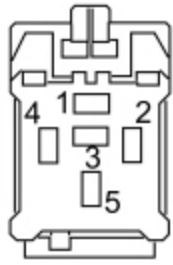
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3276 Pin 3	SBB44 (GN/RD)	-	Ground

Is the voltage greater than 11 volts?

Yes	GO to J6 .
No	VERIFY that BJB fuse 44 (15A) is OK. If OK, REPAIR the circuit for an open. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

J6 CHECK FOR GROUND TO THE POWER MIRROR FOLD RELAY

- Ignition OFF.
-



- N0149489
- Measure the **resistance** between:

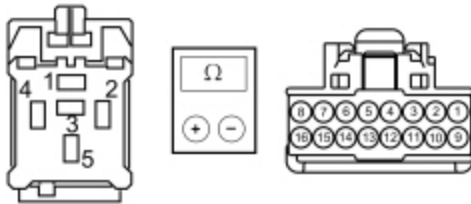
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3276 Pin 1	GD134 (BK/VT)	-	Ground

Are the resistances less than 3 ohms?

Yes	GO to J7 .
No	REPAIR the circuit. TEST the system for normal operation.

J7 CHECK THE FOLDING MIRROR OUTPUT CIRCUITS FOR AN OPEN

- Disconnect: LH Exterior Mirror [C615](#).
-



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- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3276 Pin 2	CPM04 (VT/BN)	C615 Pin 7	CPM04 (VT/BN)
C3276 Pin 5	CPM39 (GN/VT)	C615 Pin 16	CPM39 (GN/VT)

Are the resistances less than 3 ohms?

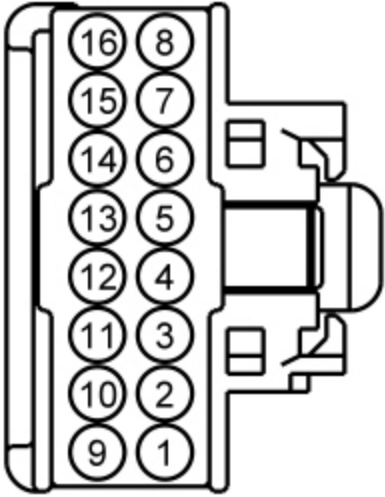
Yes	INSTALL a new power mirror fold relay. TEST the system for normal operation.
No	REPAIR the circuit(s) in question. TEST the system for normal operation.

J8 CHECK THE POWER MIRROR FOLD RELAY

- Ignition OFF.
- Disconnect: Inoperative Exterior Mirror [C521](#) or [C615](#).
- Measure the **resistance** between:

LH exterior mirror:

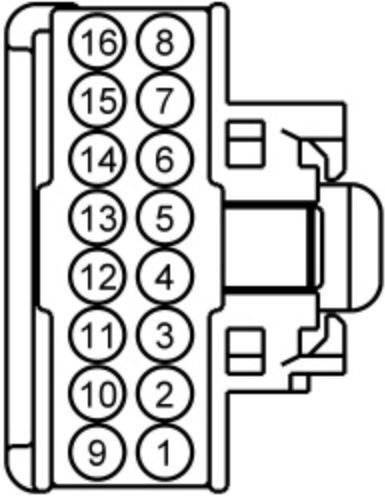
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C521 Pin 7	CPM04 (VT/BN)	-	Ground
C521 Pin 16	CPM39 (GN/VT)	-	Ground



-
- Measure the **resistance** between:

RH exterior mirror:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C615 Pin 7	CPM04 (VT/BN)	-	Ground
C615 Pin 16	CPM39 (GN/VT)	-	Ground



-

Are the resistances less than 3 ohms?

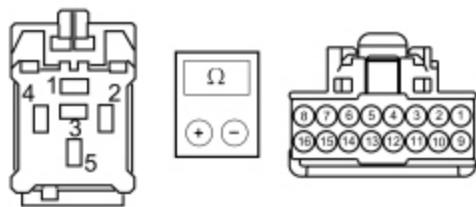
Yes	CHECK the LH or RH exterior mirror jumper harness for open circuits and damaged or pushed-out pins. If the jumper harness is not OK, REPAIR the jumper harness. If the jumper harness is OK or cannot be repaired, INSTALL a new LH or RH exterior mirror. REFER to Exterior Mirror in this section. TEST the system for normal operation.
No	GO to J9 .

J9 CHECK THE FOLDING MIRROR OUTPUT CIRCUITS FOR AN OPEN

- Disconnect: Power Mirror Fold Relay [C3276](#).
- Measure the **resistance** between:

LH exterior mirror:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3276 Pin 2	CPM04 (VT/BN)	C521 Pin 7	CPM04 (VT/BN)
C3276 Pin 5	CPM39 (GN/VT)	C521 Pin 16	CPM39 (GN/VT)

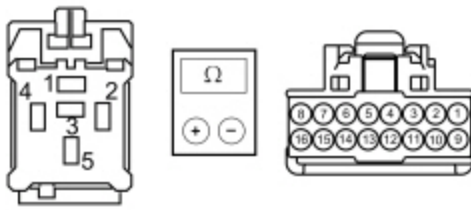


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- Measure the **resistance** between:

RH exterior mirror:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3276 Pin 2	CPM04 (VT/BN)	C615 Pin 7	CPM04 (VT/BN)
C3276 Pin 5	CPM39 (GN/VT)	C615 Pin 16	CPM39 (GN/VT)



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Are the resistances less than 3 ohms?

Yes	INSTALL a new power mirror fold relay. TEST the system for normal operation.
No	REPAIR the circuit(s) in question. TEST the system for normal operation.